KOR STRUCTURAL · ENGINEERING TOOLS

Structural models built from the drawings

31168 YMCA Langara and 31138 2170 W 1st · prepared 7 August 2026

Andrea's ETABS models were lost with a failed drive. Rather than re-enter the building's geometry by hand, this reads the concrete-outline DXF plans drafting already exports and produces an ETABS model covering every storey. It does none of the engineering — it removes the typing.

This one document covers both models. They are not equivalent: **31168 is the rebuild** — none of it existed. **31138 is a gap-fill**, because you had already rebuilt that building's walls by hand.

WHAT WAS GENERATED	31168 YMCA LANGARA	31138 2170 W 1ST
Storeys populated	**61**	23
Wall panels, true thicknesses	**918**	89
Columns, sized & oriented	**2,425**	162
Floor plates	**79**	8
Headers over openings (spandrel shells)	**141**	5
Shaft and stair openings cut	**46**	1
Pier labels	**123**	8
Members you already had, not duplicated	—	233 walls, 181 columns

What changed after your two calls

Everything below came from your feedback on 7 August and is already in the files.

YOU SAID	WHAT IT DOES NOW
"We can't have a wall go from here to here and then another one from here to here. We need a connection."	Wall centrelines form a network. Corners are carried out to where they cross and share a joint; a wall running into another splits it so the T has a joint on both members.
"We don't model them as line elements, we model them as shells… in ETABS they're called spandrels."	Headers are wall panels standing only over their opening, depth = storey height less opening height, bounded to the 24–60" range your own models use. 141 of them, labelled as spandrels.
"Every floor we have two slabs on top of each other."	Drafting issues both a range sheet and a sheet per level, so each floor arrived twice. 43 doubled plates removed.
"Below grade, the basement walls are missing."	They are drawn as two concentric rings, so each read as a building-sized rectangle and both were discarded. Now read as the one wall they are — all four sides, including the angled west one.
"These are supposed to be round columns, but they're square. Square and rotated."	A column is now modelled round only where the drawing drew it with an arc, not where its shape merely looks circular — a chamfered square passes every shape test a circle does. 31168 comes out with exactly the three diameters its drawings contain: 16" (28), 24" (19) and 30" (348), and none besides.
"It filled the volume, but didn't do the actual corner."	A pier and an L of two walls both fill their bounding box, so fill could not separate them. Width where the outline has substance can: piers stay whole, corners decompose into the two walls that are there.
"The lowest level, which is P3, seems way too high."	ETABS parks the base far below the building and folds the whole distance into the lowest storey — 31168 exported with the base at −1,000 ft and LEVEL P3 1,113 ft tall, so on import every member down there was extruded that far and the parkade arrived as a solid block half the height of the building. The storey list itself is now rewritten: the lowest storey takes a typical height from its own building and the base rises by exactly as much, so every other elevation is untouched. 31168 spans 104–561 ft, with the roof still at 561.1 ft and the lowest slab still at 113.9 ft.
"Less than 48 in length should be a column" — then "this should be a wall" for core faces	Connection decides, not length: under 48" and joined to no other wall is a column; a short face forming part of a core stays a wall.
Loads, diaphragms, stiffness modifiers,	Removed. No diaphragms are assigned, no loads applied.

section properties are yours

What to open

Each sits in its project's `02 Engineering\02 Lateral Design\01 ETABS Models\`. In ETABS: **File** → **Import** → **ETABS .e2k Text File**.

FILE	WHAT IT IS
`31168-FROM-DRAWINGS.e2k`	**The one that matters.** Tower B and the rest of the building — 60 storeys, 897 walls, 2,418 columns, 124 plates. Nothing of this existed before.
`31138-FROM-DRAWINGS.e2k`	A gap-fill. You have already rebuilt this building's walls — 23 to 28 panels on every storey — so this adds only what the drawings show and your model does not: 98 walls and 118 columns, mostly at podium and parkade. 233 walls and 181 columns of yours were recognised at those locations and deliberately not modelled again.
`…-report.txt`	Every sheet, the storeys it landed on, what it produced, and every location that needed judgement.
`…-MODEL-VIEWS.png`	The finished model drawn in isometric, two elevations and plan — what you will see when you import it. Worth a glance before you open ETABS.
`_drawing-renders\`	The drawings themselves, with the extracted members drawn over them. Quickest way to see what was read and what was not.

Everything generated is named `KW*`, `KC*`, `KP*` with new sections prefixed `KOR-`, so you can select, filter or delete the whole lot in one action. Where the model already had a section of the right thickness it was reused, so members carry the project's own concrete and names you recognise.

What has been verified, and what has not

Verified: the model is drawn and looked at before it is sent

Counts and coordinates cannot see the faults that make a model wrong at a glance — a wall two inches tall passes every count. So the generator now renders its own output to an image (isometric, two elevations, plan) and that image is checked before the file goes anywhere; it ships beside the model as `…-MODEL-VIEWS.png`. Three faults were found this way and fixed, none of which any numeric check had noticed:

- **Joints carried elevations.** An ETABS joint is a plan position only — the third number on a `POINT` line is an offset from the storey the member is assigned to, and elevation comes from that storey. Writing a true elevation there throws a member hundreds of feet off its storey. Your own 31138 model settles it: 1,098 of its 1,156 joints carry nothing but X and Y. Members are now written the way ETABS writes them — a wall as its two plan joints repeated (`PANEL 4 "a" "b" "b" "a" 1 1 0 0`), a column as one joint rising a storey.

- **Towers were built as wafers.** 31168's storey list is the whole site, so it holds a storey for every tower's floor: because tower B's level 34 sits 2.4″ above tower A's, the export contains a storey named `B-LEVEL 34` that is two inches tall. Taking a storey's own height as its wall height made 78 of 897 panels two-inch wafers hanging a storey above their floor. A wall now stands on the previous floor *of its own tower*, and is assigned to every storey it passes through so ETABS builds it as one continuous run. The towers interleave at 4.67 ft and 5.0 ft, so no numeric threshold could have separated a real storey from a duplicate floor — only the storey name can.
- **Scraps of floor hung in space.** Slab-edge linework closes into small rings that are not floors. Seven of 31138's fourteen plates were 52–68 ft² against a real tower floor of 9,666 ft². Measured across both projects the two populations do not overlap — standalone rings come out at 52–115 ft², real plates at 915 ft² and up — so a ring must now reach 400 ft² to be modelled as a plate.

Those three are now permanent checks, run against both real projects on every build: no wall or column shorter than a person, none taller than a double-height storey, no plate smaller than a room, and no storey in the reference that is not storey-height. 410 tests pass.

Verified: every member sits exactly where the drawing draws it

Each plan is rendered with the extracted members drawn over it (`takeoff dxf-render <plan> <png> --overlay`), so what the model carries can be compared against the linework directly rather than inferred. On B-LEVEL 30 the core's two shafts, its bar wall and all twenty-four perimeter columns land on their outlines; the images are in `_drawing-renders\` . Wall thicknesses and column sizes are measured from the outlines, not assumed.

Verified: against your own model

On 31138 the generated columns land where you placed yours by hand: no shift applied, 43 of your 87 columns matched within six inches, and on L10 the median distance was 0.0. On 31168 the wall inventory agrees with the model that was lost — its pier-force export records five wall piers on every typical storey and 34 at L01, and the drawings give five core elements per typical storey and 29 walls at L01.

Verified: the geometry lands on your grid

The drawings and the model both come out of the same Revit project, so they already share a coordinate system and the geometry is placed without any shift. That is checked against the grid itself rather than assumed: the tower core straddles grids 15 and 16, which ETABS places at x = 1500 and 1826, and the walls read out of the drawing sit at 1500.05 and 1826.05 — a twentieth of an inch. The same holds up the tower, with wall coordinates landing on grid lines on B-LEVEL 30, B-LEVEL 33 and LEVEL 20.

Worth knowing when you look: only the inner face of a core wall sits on its grid line, the outer face standing off by the wall thickness, which is how the drawing draws it.

What it reads, and how

LAYER	BECOMES	HOW
`JBP_V-WALL`	Wall panels	A core is drawn as one ribbon tracing both faces of a group of walls. Each face is paired with the face opposite it; the pair gives the centreline and the true thickness.
`JBP_V_COL`	Columns	Footprint gives size and rotation, so a 16×24 sits the way it is drawn.
`JBP_C_SLABEDG*`	Floor plates	Outlines stitched into rings; rings inside a ring are openings.

Sheet titles carry the placement. `LEVEL 29 PLAN (L29-35)` fills seven storeys; `A-LEVEL 28` goes to Tower A alone; `LEVEL P2` to the parkade. The geometry is merged into an `.e2k` that ETABS itself exported, so storeys, grids, materials and design settings remain ETABS's own — the file differs from a known-good one only by the lines added.

Limitations — read before you rely on it

Slab plates are the weak half

Drafting interrupts slab edges wherever other linework crosses them, so many outlines do not close. Those are dropped and reported rather than guessed at, and a standalone ring must reach 400 ft² to be modelled — the 124 plates that survive have a median area of 9,800 ft², which is a real floor plate rather than a sliver. Coverage is therefore partial: the plates that are there are believable, but not every storey has one. Treat them as a starting point, or ignore them and place your own diaphragms.

Some storeys carry walls and columns but no plate, because their slab edges will not close. On **31168** that is seven — `LEVEL P1`, `P2`, `P3`, `LEVEL 1 MEZZ`, `C-LEVEL 3`, `B-LEVEL 28`, `A-LEVEL 35`. On **31138** it is twelve — `L01` – `L06`, `L08`, `L11`, `L14`, `L16`, `L17`, `L20`, though there your own plates already cover most of them. Each list is printed at the top of that project's report and in its questionnaire. Those storeys have no diaphragm until a plate is drawn there, and in a 3D view their members are the ones that look unsupported. This is the largest remaining gap, and it is a drawing-data limit rather than a rule that can be tuned: widening the closure tolerance was tried and measured, and every plate it added was a fragment.

Widening the closure tolerance was tried and rejected on the evidence: at 18″ the model gained 302 more plates, but the count of plates in a plausible size range did not change at all — every addition was a fragment. Recovering the missing plates needs a better method, not a looser one (see below), and that is why only one model is shipped.

What is set up, and what is yours to decide

The model arrives load-ready rather than load-less. Load patterns (Dead with self-weight, Live), load cases (Modal at 55 modes, Dead, Live) and the mass source come through from the reference, and because every wall, column and plate now carries its true thickness and the project's own concrete, **self-weight is real the moment you run it**. On 31138 your own 149 area-load assignments are preserved untouched.

What is not applied is superimposed dead and live load on the generated plates, and that is deliberate. Your own 31138 model carries five different SDL values on L01 alone — 5, 25, 45, 70 and 145 psf — with live load from −50 to 150, because they follow occupancy: terraces, planters, amenity, suites, corridors. None of that is visible in a structural outline, and putting a storey's "typical" value on a landscaped terrace would flow straight through to the foundation loads.

A rigid diaphragm per storey is assigned to every generated plate, so modal analysis runs meaningfully out of the box. Also yours: spectra, stiffness modifiers, pier and spandrel labels, and meshing.

Openings are detected but not cut

Shafts and stair openings are found as inner rings and reported, but not subtracted from the plates, so slab areas read high where a core passes through.

> **Check the thick walls.** Anything modelled thicker than 24″ is flagged in the report. Wall outlines that meet at junctions can pair across the junction and read thicker than the wall really is. The flags tell you exactly where to look.

31168's storeys are the whole site, not Tower B alone

The model that was lost was Tower B on its own, storeys L01 to L40 with Mezz and P1–P2. This one is built on the site model exported from Revit, so its storeys are the whole development — `B-LEVEL 27` upward for the tower, shared unprefixed storeys lower down. The geometry is right; the frame it sits in is not the one you were working in. Worth deciding before you build on it.

It believes the drawing

Thicknesses and sizes come from what was drawn, not from a schedule. A wall drawn at the wrong width models at the wrong width. The plans used here are dated 28 June 2026 — later revisions are not in this model.

The questions, in one place

Beside each model is `…-QUESTIONS-for-Andrea.xlsx`. It lists every decision the tool had to make without being told — wall or pier, doorway or end of concrete, which storeys need plates, what superimposed load belongs where — with what it did, why it matters, and the evidence behind it. There is an answer column.

Those answers are the point. Each one becomes a rule the tool applies from then on, on this project and every one after, instead of a judgement folded into the code where nobody can see it. The workbook also carries every flagged location and a ledger of which drawing filled which storey, so a wrong answer is traceable to the sheet that caused it.

What would make it materially better — your input decides

IMPROVEMENT	WHAT IT NEEDS FROM YOU
Cut the openings. Subtract shafts and stairs from the plates so slab areas and diaphragms are right.	Confirm that inner rings on the slab-edge layers are always openings, never a step or a depression.
Close slab edges properly by extending an interrupted edge along its own direction to the line that cut it, instead of bridging by distance. Measurement says this is the only way to recover the missing plates.	Nothing — but tell us which storeys you most need plates on, so it is proven where it matters.
Wall grades by height. Read the wall schedule so 65 MPa low and 45 MPa high come through instead of one material.	Point us at the schedule sheet you would read for this, and how you would expect the grades keyed to walls.
Beams. Nothing on the beam layers is read today.	Whether beams are worth having, or you would rather add them yourself.
A button in the KOR app instead of a command line.	Whether you would run this yourself per project, or ask for it.

The single most useful thing you can send back: open the model, and tell us the first three things that are wrong or annoying. Each one is a rule the tool does not yet know.

Where this sits in the KOR toolset

This is a new front end on the engineering engine the app already carries, not a separate program.

```
Revit  ──→  drafting's DXF plan exports  ──→  [ DXF → ETABS ]  ──→  .e2k  ──→  ETABS
PDF drawings ──────────────────────────────→ [ PdfToSafe   ]  ──→  .f2k / .e2k / live OAPI
PDF drawings ──────────────────────────────→ [ Takeoff     ]  ──→  quantities, xlsx
```

All three live in `Kor.Operations.EngineeringTools`, share the same geometry code, and are covered by the same test suite (384 tests). PdfToSafe reads drawings when only a PDF exists; this reads the DXFs when drafting has exported them, which is exact and needs no vision model. The natural next step is a button in the app's Engineering Tools panel so it runs without a command line.

> **It only stays reliable while the layer names hold.** `JBP_V-WALL`, `JBP_V_COL` and `JBP_C_SLABEDG` are now a machine contract, not just a drafting habit — which is a concrete argument for the Revit template and standards work already under way.

Built 6 August 2026 · `Kor.Operations.EngineeringTools.Core/Dxf` · CLI `takeoff dxf-to-etabs`, `dxf-inspect`, `e2k-compare` · documented at `docs/DxfToEtabs.md`